Random Sampling

STAT 3160: Applied Multivariate Analysis

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Random Samples

In order to study the sampling variability of statistics like $\bar{\mathbf{x}}$ and $\mathbf{S}$ with the ultimate aim of making inferences, we need to make assumptions. Given the random matrix

$$\mathbf{X} = \begin{pmatrix} X_{11} & X_{12} & \dots & X_{1p} \\ X_{21} & X_{22} & \dots & X_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ X_{n1} & X_{n2} & \dots & X_{np} \end{pmatrix} = \begin{pmatrix} X_1' \\ X_2' \\ \vdots \\ X_n' \end{pmatrix}.$$

- Each row of the data is unrelated to all the others (i.e., $X_1, X_2, \dots, X_n$ are all independent).
  1. Each row corresponds to a case, individual, sampling unit.
  2. Each row is a multivariate observation in p-dimensional space.
- The joint distribution of all p variables is the same for all cases (i.e., $X_1, X_2, \dots, X_n$ have the same distribution).

Overall Measures of Dispersion

Sometimes it is also useful to have an overall measure of dispersion in the data. In this measure, it would be good to include all of the variables simultaneously, rather than one at a time. In the past, we looked at the individual variables and their variances to measure the individual variances. Here we are going to look at measures of dispersion of all variables together, particularly we are going to look at such measures that look at the total variation.

1. Generalized Variance
2. Total Variation

Generalized Variance

By definition, the generalized variance of a random vector $\mathbf{X}$ is equal to $|\Sigma|$, the determinant of the variance/covariance matrix. The generalized variance can be estimated by calculating $|\mathbf{S}|$, the determinant of the sample variance/covariance matrix.

GSV: p-space interpretation

Define the sample mean vector $\bar{\mathbf{x}}' = (\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_p)$. Consider the following equation:

$$(\mathbf{x} - \bar{\mathbf{x}})'\mathbf{S}^{-1}(\mathbf{x} - \bar{\mathbf{x}}) = c^2.$$

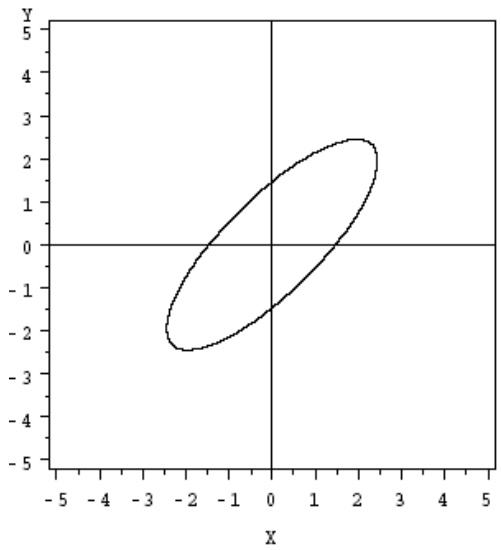
The above equation defines a hyperellipsoid centered at $\bar{\mathbf{x}}$. For example, if $p = 2$, one has

$$a_{11}(x_1 - \bar{x}_1)^2 + 2a_{12}(x_1 - \bar{x}_1)(x_2 - \bar{x}_2) + a_{22}(x_2 - \bar{x}_2)^2 = c^2,$$

where a_{11}, a_{12} and a_{22} depend on

$$\mathbf{S} = \begin{pmatrix} s_{11} & s_{12} \\ s_{21} & s_{22} \end{pmatrix} \longrightarrow \mathbf{S}^{-1} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

GSV Interpretation



The volume of this ellipsoid is

$$\text{volume}\{\mathbf{x} : (\mathbf{x} - \bar{\mathbf{x}})'\mathbf{S}^{-1}(\mathbf{x} - \bar{\mathbf{x}}) \leq c^2\} = (\text{constant})|\mathbf{S}|^{1/2}c^p.$$

Note: $|\mathbf{S}| = GSV$.

Implication & Limitation of GSV

- GSV is zero if and only if the deviation vectors $\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_p$ are linearly dependent.
- The GSV represents both variability and covariability.
- You can have very different patterns of variability and association (correlation) but have the same value of GSV.
- Example:

$$\mathbf{S}_0 = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \quad \rightarrow \quad |\mathbf{S}_0| = 9$$

$$\mathbf{S}_1 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} \quad \rightarrow \quad |\mathbf{S}_1| = 9$$

$$\mathbf{S}_3 = \begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix} \quad \rightarrow \quad |\mathbf{S}_3| = 9$$

Another Global Statistic

Another overall measure of dispersion is **Total Variation**. The total variation of a random vector X is simply the trace of the population variance-covariance matrix.

$$\text{Total variance} = trace(\Sigma) = \sigma_1^2 + \sigma_2^2 + \ldots + \sigma_p^2.$$

The total variation can be estimated by:

$$\text{Total sample variance} = trace(\mathbf{S}) = s_1^2 + s_2^2 + \ldots + s_p^2.$$

The total variation is of interest for principal components analysis and factor analysis and we will look at these concepts later in this course.

Example: Women's Nutrition Data

The generalized variance for the Women's Nutrition data can be calculated using the SAS.

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Example: Nutrient Intake Data - Generalized Variance

        S                                                    GENVAR

157829.44 940.08944 6075.8163 102411.13  6701.616  2.831E19
940.08944 35.810536 114.05803 2383.1534 137.67199
6075.8163 114.05803 934.87688 7330.0515 477.19978
102411.13 2383.1534 7330.0515 2668452.4 22063.249
 6701.616 137.67199 477.19978 22063.249 5416.2641
```

We have the exact same numbers that were presented before. Here we see that the generalized variance is:

$$|\mathbf{S}| = 2.83 \times 10^{19}$$

This represents a very large volume of space. Again, the interpretation of this particular number depends largely on subject matter knowledge. In this case, we can not say if this is a particularly large number or not unless we know more about women's nutrition.

Linear Combination of Variables

- A **linear combination** of p variables $X_1, X_2, \ldots, X_p$ is $a_1X_1 + a_2X_2 + \ldots + a_pX_p$. Let

$$\mathbf{a}' = (a_1, a_2, \ldots, a_p) \text{ and } X = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{pmatrix}$$

  A linear combination in terms of vector operations is $\mathbf{a}'X$.
- The linear combination $\mathbf{a}'X$ is itself a random variable.
- Let μ and $\mathbf{\Sigma}$ denote the mean vector and variance-covariance matrix for random vector X. One has

$$E(\mathbf{a}'X) = \mathbf{a}'\mu \text{ and } \text{var}(\mathbf{a}'X) = \mathbf{a}'\mathbf{\Sigma}\mathbf{a}$$

Linear Combination of Variables

If we add a constant to the linear combination,

$$a_1X_1 + a_2X_2 + \ldots + a_pX_p + c = \mathbf{a}'X + c$$

Then

- Mean: Change location

$$E(\mathbf{a}'X + c) = \mathbf{a}'E(X) + c = \mathbf{a}'\mu + c$$

- Variance: No change

$$\text{var}(\mathbf{a}'X + c) = \mathbf{a}'\boldsymbol{\Sigma}\mathbf{a}$$

Multiple Linear Combination

Sometimes one just isn't enough...

Suppose that there are q linear combinations of the p random variables

$$\begin{aligned} y_1 &= a_{11}X_1 + a_{12}X_2 + \ldots + a_{1p}X_p \\ y_2 &= a_{21}X_1 + a_{22}X_2 + \ldots + a_{2p}X_p \\ &\vdots \\ y_q &= a_{q1}X_1 + a_{q2}X_2 + \ldots + a_{qp}X_p \end{aligned}$$

In matrix form

$$\mathbf{y}_{q\times 1} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_q \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1p} \\ a_{21} & a_{22} & \ldots & a_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ a_{q1} & a_{q2} & \ldots & a_{qp} \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{pmatrix} = \mathbf{A}_{q\times p} X_{p\times 1}$$

No restriction on q, but usually $q \leq p$.

Mean and Covariance for q

$$\mathbf{y}_{q\times 1} = \mathbf{A}_{q\times p} X_{p\times 1}.$$

Then

- Mean:

$$\mu_{\mathbf{y}} = E(\mathbf{y}) = E(\mathbf{A}X) = \mathbf{A}E(X) = \mathbf{A}\mu_{\mathbf{x}}$$

- Variance-covariance matrix

$$\mathbf{\Sigma}_{\mathbf{y}} = \mathrm{cov}(\mathbf{y}) = \mathrm{cov}(\mathbf{A}X) = \mathbf{A}\mathbf{\Sigma}\mathbf{A}'$$

Two sets of linear combinations

If we have one set of q linear combinations and another set of r linear combinations of random variables X, then

$$Y_{q\times 1} = \mathbf{A}_{q\times p} X_{p\times 1}, \quad E(Y) = \mathbf{A}\mu_{\mathbf{x}}, \quad \mathbf{\Sigma_y} = \mathbf{A}\mathbf{\Sigma_x}\mathbf{A}'$$

$$Z_{r\times 1} = \mathbf{B}_{r\times p} X_{p\times 1}, \quad E(Z) = \mathbf{B}\mu_{\mathbf{x}}, \quad \mathbf{\Sigma_z} = \mathbf{B}\mathbf{\Sigma_x}\mathbf{B}'$$

and

$$\mathbf{\Sigma_{yz}} = \mathbf{A}\mathbf{\Sigma_x}\mathbf{B}'$$

Example

Let $X' = (X_1, X_2)$ be a random vector with mean vector $\mu'_{\mathbf{x}} = (\mu_1, \mu_2)$ and

$$\Sigma_x = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix}.$$

Find the mean vector and covariance matrix for the linear combinations

$$\begin{aligned} Z_1 &= X_1 - X_2 \\ Z_2 &= X_1 + X_2 \end{aligned}$$

Matrix operation for sample mean and covariance

Given a sample data

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}.$$

The sample mean vector $\bar{\mathbf{x}}$

$$\bar{\mathbf{x}} = \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_p \end{pmatrix} = \frac{1}{n} \begin{pmatrix} x_{11} & x_{21} & \cdots & x_{n1} \\ x_{12} & x_{22} & \cdots & x_{n2} \\ \vdots & \vdots & \vdots & \vdots \\ x_{1p} & x_{2p} & \cdots & x_{np} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$$

or

$$\bar{\mathbf{x}} = \frac{1}{n}\mathbf{X}'\mathbf{1}$$

One can also show that

$$\mathbf{S} = \frac{1}{n-1}\mathbf{X}'\left(\mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}'\right)\mathbf{X}$$